

Lecture# 10

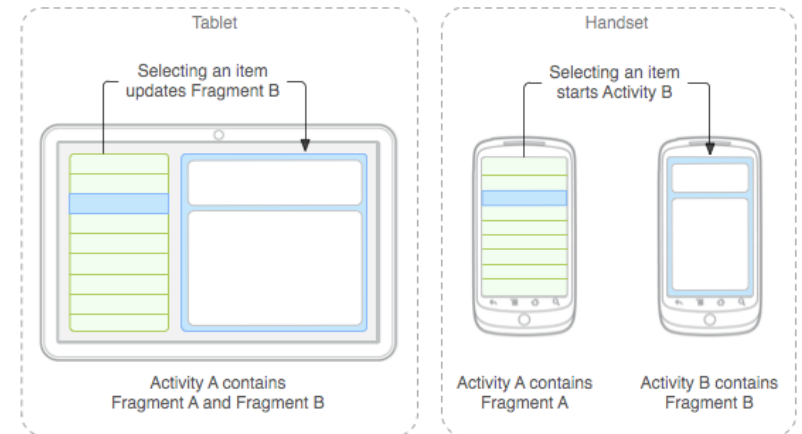
Fragments

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Fragments

- A **Fragment** is a piece of an activity which enable more modular activity design
- A fragment is a kind of **sub-activity**
- A **Fragment** represents a behaviour or a portion of user interface in a **FragmentActivity**
- **Fragment** can be used to display different screen layout for different screen size devices e.g. mobile or tablet



Feature of Fragments

- A fragment has its own layout
- You can add or remove fragments in an activity while the activity is running
- You can combine multiple fragments in a single activity to build a multi-pane UI
- A fragment can be used in multiple activities

Lifecycle of Android Fragment

- `onCreateView()`
 - To draw a UI for your fragment, you must return a view from this method that is the root of your fragment's layout.

Creating Fragment

- To create a fragment, you must create a subclass of “Fragment”
- The fragment class has code that looks a lot like an “Activity”

```
import androidx.fragment.app.Fragment;  
  
public class FragmentA extends Fragment {  
}
```

Adding a User Interface

- A fragment is usually used as part of an activity's user interface and contributes its own layout to the activity.
- To provide a layout for a fragment, you must implement the `onCreateView()` method, which the Android system calls when it's time for the fragment to draw its layout.
- Your implementation of this method must return a view that is the root of your fragment's layout.

Adding a User Interface (Cont.)

- 1st parameter: The ViewGroup to be the parent of the inflated layout.
- 2nd parameter: Passing the container is important in order for the system to apply layout parameters to the root view of the inflated layout, specified by the parent view in which it's going.
- 3rd parameter: A boolean indicating whether the inflated layout should be attached to the ViewGroup during inflation

```
@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container,
                          Bundle savedInstanceState) {
    // Inflate the layout for this fragment
    return inflater.inflate(R.layout.fragment_layout, container, false);
}
```

Adding Fragment to Activity

- Consider a simple example for replacing two different fragment in the main activity
- To add fragment programmatically in the code, we need fragment manger object from “FragmentManager” class using this

```
FragmentManager fragmentManager = getSupportFragmentManager();
```

- To replace fragment from the main activity class, fragment transaction will be need to perform transaction between fragments.

```
FragmentTransaction fragmentManager.beginTransaction();
```

Adding Fragment to Activity (Cont.)

- Now for adding or replacing new fragment in the main activity, we need to add or replace new fragment in the fragment transaction object and perform transaction using “commit()” method in fragment transaction class
- The new fragment must be the object from the fragment class

```
FragmentA fragmentA = new FragmentA();  
fragmentTransaction.replace(R.id.fragment_container, fragmentA);  
fragmentTransaction.addToBackStack(null);  
fragmentTransaction.commit();
```

- The “addToBackStack(null)” adds the transaction to a back stack of fragment transactions
- The fragment container is place holder for fragment in the main activity XML layout

Example 1: Replacing Fragment

fragment_a.xml

```
<?xml version="1.0" encoding="utf-8" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/green"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="Fragment A" />

</LinearLayout>
```

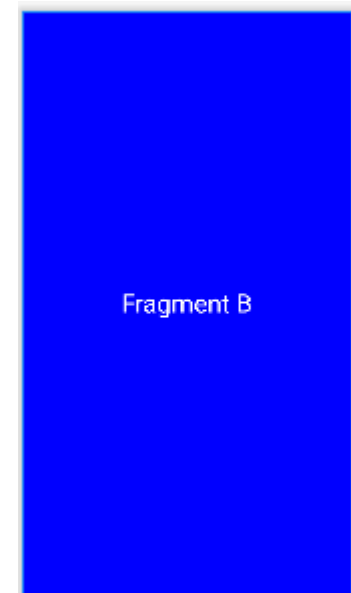


fragment_b.xml

```
<?xml version="1.0" encoding="utf-8" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/blue"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:textSize="32sp"
        android:gravity="center"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="Fragment B" />

</LinearLayout>
```



Create Main Activity XML Layout

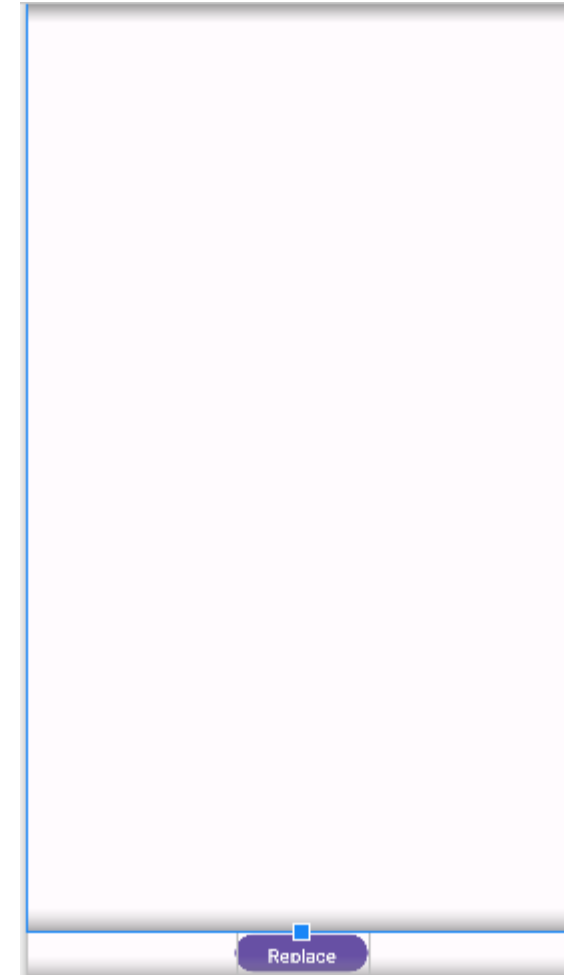
main_activity.xml

```
<?xml version="1.0" encoding="utf-8" ?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:weightSum="4"
    tools:context="Fragments.MainFragmentActivity">

    <LinearLayout
        android:orientation="vertical"
        android:id="@+id/frame_container"
        android:layout_weight="3"
        android:layout_width="match_parent"
        android:layout_height="match_parent">

    </LinearLayout>

    <Button
        android:id="@+id/replace_btn"
        android:layout_weight="1"
        android:layout_gravity="center"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Replace" />
</LinearLayout>
```



Create Two Fragments

FragmentA.java

```
public class FragmentA extends Fragment {  
    @Override  
    public View onCreateView(LayoutInflater inflater, ViewGroup container,  
                             Bundle savedInstanceState) {  
        // Inflate the layout for this fragment  
        return inflater.inflate(R.layout.fragment_a, container, false);  
    }  
}
```

FragmentB.java

```
public class FragmentB extends Fragment {  
    @Override  
    public View onCreateView(LayoutInflater inflater, ViewGroup container,  
                             Bundle savedInstanceState) {  
        // Inflate the layout for this fragment  
        return inflater.inflate(R.layout.fragment_b, container, false);  
    }  
}
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    boolean isFragmentA = true;
    FragmentManager fragmentManager;
    FragmentTransaction fragmentTransaction;
    FragmentA fragmentA;
    FragmentB fragmentB;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        fragmentA = new FragmentA();
        fragmentB = new FragmentB();

        replaceFragment(fragmentA);

        Button replaceBtn = findViewById(R.id.replace_btn);
        replaceBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if(!isFragmentA) {
                    replaceFragment(fragmentA);
                    isFragmentA = true;
                } else {
                    replaceFragment(fragmentB);
                    isFragmentA = false;
                }
            }
        });
    }

    void replaceFragment(Fragment fragment){
        fragmentManager = getSupportFragmentManager();
        fragmentTransaction = fragmentManager.beginTransaction();
        fragmentTransaction.replace(R.id.fragment_container, fragment);
        fragmentTransaction.addToBackStack(null);
        fragmentTransaction.commit();
    }
}
```



Pass Data Between Fragment & Activity

- The data can be pass between activity and fragment using “Bundle” class.

```
FragmentA fragmentA = new FragmentA();  
Bundle args = new Bundle();  
  
args.putString("arg_key", "This is string from arguments");  
fragmentA.setArguments(args);
```

- In fragment “onCreateView()” method the arguments can be received

```
@Override  
public View onCreateView(LayoutInflater inflater, ViewGroup container,  
                          Bundle savedInstanceState) {  
    if(getArguments() != null) {  
        String string = getArguments().getString("arg_key");  
        Log.d("Fragment A", string);  
    }  
    // Inflate the layout for this fragment  
    return inflater.inflate(R.layout.fragment_first, container, false);  
}
```